CROSSSAFE · DESIGN DECISION RECORD

CrossSafe concept selection

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DOCUMENT CONTROL

Document ID	CRS-DDR-004	Version	0.1
Status	Draft	Date	2026-10-31
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/crosssafe		
Source	docs/decisions/0004-concept-selection.md @ d9f28be		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Concept options A, B and C scored against CRS-URS-001 (kit 1.9.1 upgrade); A and C within 10 %, both to Amish; proposed, awaiting Amish

0004: Concept selection

- **Date:** 2026-10-31
- **Status:** proposed, awaiting Amish. Option A continues unchanged until he decides.

Context

Kit 1.8 asks every repo for at least two concept options with different working principles, scored against the user requirements. CrossSafe was designed as one concept (the pole-top beacon). It is option A and is not redesigned. Two options with other working principles were added and all three were modelled to D3 (`cad/fidelity.json`): B moves the warning to road level with a kerbside bollard and solar studs, and C removes the electronics with a pull-up retroreflective panel. The full numbers and the scoring are in CRS-OPT-001.

Options considered

Table 1. Options and weighted scores (maximum 185).

Option	Principle	Parts	Mass	Cost (USD)	Score	Share
A	Pole-top flashing beacon with detection	117	27.6 kg	364.00 (BOM)	138	74.6 %
B	Kerbside solar bollard with in-road studs	156	45.5 kg	about 400 (estimate)	108	58.4 %
C	Pull-up retroreflective panel, no electricity	40	10.9 kg	about 70 (estimate)	128	69.2 %

Decision

Proposed, awaiting Amish: A and C are within 10 % (C is 7.2 % lower), so both go to him without a pick. The author leans to A, because it alone detects, confirms and warns on both sides, which is what the evidence of effect (FHWA) rests on, and because it is the design all TRL 3 work covers. C is the better answer for a crossing with no budget, and B is not recommended. No document, drawing or BOM changes if Amish keeps A. If he picks C, the BOM, build plan and calculations are redone for C at TRL 3, and the repo's pitch changes from a detecting beacon to a low-cost warning aid (a change of pitch, which is his decision).

Consequences

`project.yaml` keeps `concept_chosen` unset and lists the options. The scoring is the author's judgment and sensitive to the weight of cost (A leads at every weight; the gap is 12 % at weight 1 and 5 % at weight 4). The costs of B and C are estimates by analogy with the BOM of A. Co-design would test the scores for UR-001 and UR-006 first.